Document Classification using NLP Techniques

Machine Learning Project

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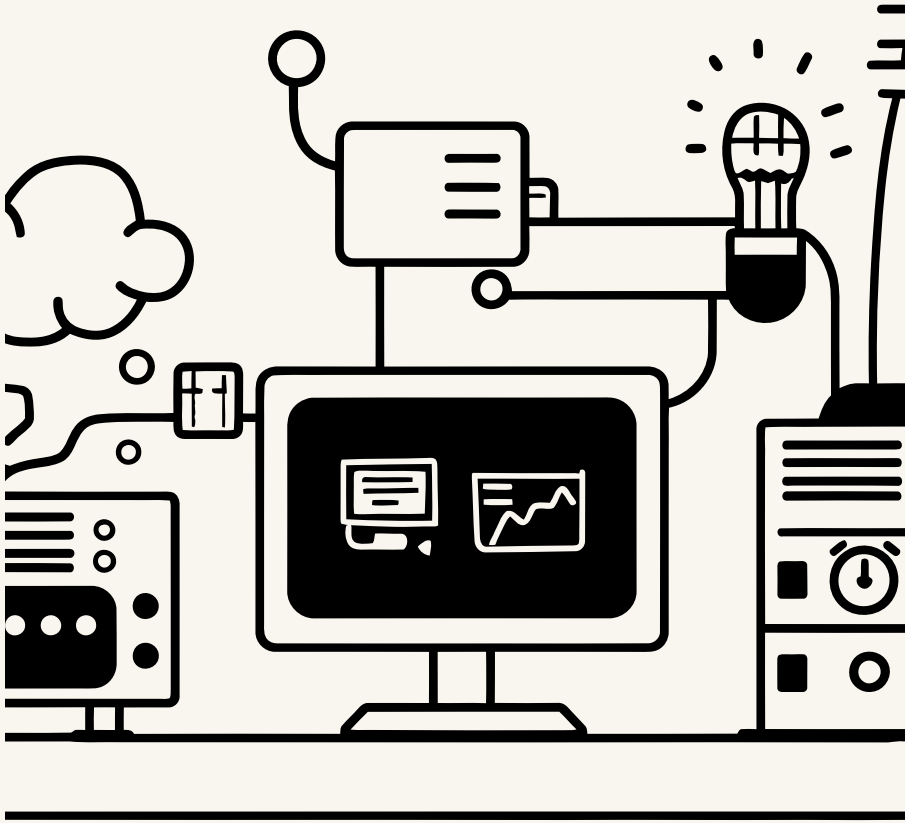

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What is Natural Language Processing?

Natural Language Processing (NLP) is a branch of artificial intelligence that helps computers understand, interpret, and generate human language. Think of it as teaching machines to communicate with us in our own language rather than requiring us to learn computer code.

Text classification is one of the most practical applications of NLP. It's like teaching a computer to sort your emails, news articles, or documents into categories automatically. Imagine having thousands of documents and needing to organize them - manual sorting would take forever, but an NLP system can do it in seconds!

News Categorization

Sorting articles into topics like sports, business, or entertainment

Spam Detection

Identifying unwanted emails automatically

Customer Support

Routing customer queries to the right department

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The Problem We're Solving

Why Manual Classification Doesn't Work

When you have hundreds or thousands of text documents, manually reading and categorizing each one becomes incredibly time-consuming and error-prone. It's like trying to sort a massive pile of papers by hand - you'll get tired, make mistakes, and waste valuable time.

Our Goal

Build an automated system that can predict the correct category of any text document with high accuracy. The computer should learn from examples and then classify new, unseen documents correctly.

1. **Collect Documents**
   Gather text data
2. **Preprocess Text**
   Clean and prepare data
3. **Train Model**
   Teach the algorithm
4. **Classify Automatically**
   Predict categories

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The BBC Text Dataset

We used a popular dataset containing 2,225 text documents from BBC news articles. This dataset is perfect for text classification because it's clean, well-organized, and covers multiple categories.

Dataset Overview

2,225 documents in total, divided equally among five distinct categories. Each document is a news article with its content and corresponding category label.

Five Categories

Business, Entertainment, Politics, Sport, and Tech - covering major news domains that require different expertise

Real-World Data

Articles are written in natural language, making this a realistic challenge rather than a simplified toy example

2,225

Total Documents

News articles in the dataset

5

Categories

Distinct topics to classify

~500

Avg per Category

Documents per topic

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Python Libraries & Tools

We used several powerful Python libraries to build our document classification system. Each library has a specific purpose in our pipeline.

Python

The main programming language that ties everything together. Known for its simplicity and powerful libraries for machine learning.

Pandas

Handles our data like a spreadsheet in code. Loads the dataset, organizes it into tables, and makes it easy to manipulate and analyze.

NLTK

Natural Language Toolkit - helps us clean and prepare text by removing stopwords (common words like "the", "and") and tokenizing (splitting text into words).

Scikit-learn

Provides the machine learning algorithms. We use it for TF-IDF vectorization and Logistic Regression. It's like having a toolbox of ML techniques ready to use.

Matplotlib & Seaborn

Create visualizations to understand our data and model performance. Charts help us see patterns and evaluate how well our classifier works.

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Data Preprocessing: Cleaning the Text

Raw text data is messy! Before we can feed it to a machine learning model, we need to clean and standardize it. Think of this as washing and preparing vegetables before cooking - the quality of our final dish depends on proper preparation.

Convert to Lowercase

Change all text to lowercase so "Hello" and "hello" are treated the same. This prevents the model from thinking they're different words.

Remove Punctuation

Get rid of commas, periods, exclamation marks, and other punctuation. These don't help with meaning but can confuse the model.

Remove Stopwords

Filter out common words like "the", "is", "and", "in" that don't carry much meaning. NLTK provides a list of these words automatically.

Tokenization

Split text into individual words (tokens). For example, "Hello world!" becomes ["hello", "world"]. This creates the basic units our model works with.

Example Transformation

Original: "The Quick Brown Fox Jumps Over the Lazy Dog!"

Processed: ["quick", "brown", "fox", "jumps", "over", "lazy", "dog"]

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Feature Extraction: TF-IDF Vectorization

Machine learning models can't work directly with text - they need numbers! TF-IDF (Term Frequency-Inverse Document Frequency) converts text into numerical vectors that capture the importance of words.

What is TF-IDF?

Term Frequency (TF): How often a word appears in a document. Words that appear frequently are probably important to that document's meaning.

Inverse Document Frequency (IDF): How rare a word is across all documents. Words that appear in many documents (like "the") are less valuable for classification than rare words.

Why TF-IDF Works

TF-IDF gives high scores to words that are frequent in a specific document but rare across all documents. This helps identify words that are characteristic of particular categories.

For example, "football" might appear frequently in sports articles but rarely in business articles, making it a good classifier for the sports category.

1

Calculate Term Frequency

Count word occurrences in each document

2

Calculate Inverse Document Frequency

Measure word rarity across all documents

3

Compute TF-IDF Score

Multiply TF by IDF for each word

4

Create Document Vectors

Combine all word scores into numerical representation

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Logistic Regression Model

Logistic Regression is a powerful and interpretable algorithm for classification tasks. Despite its name, it's used for classification, not regression! It works by finding the best line (or hyperplane in higher dimensions) that separates different categories.

Train-Test Split

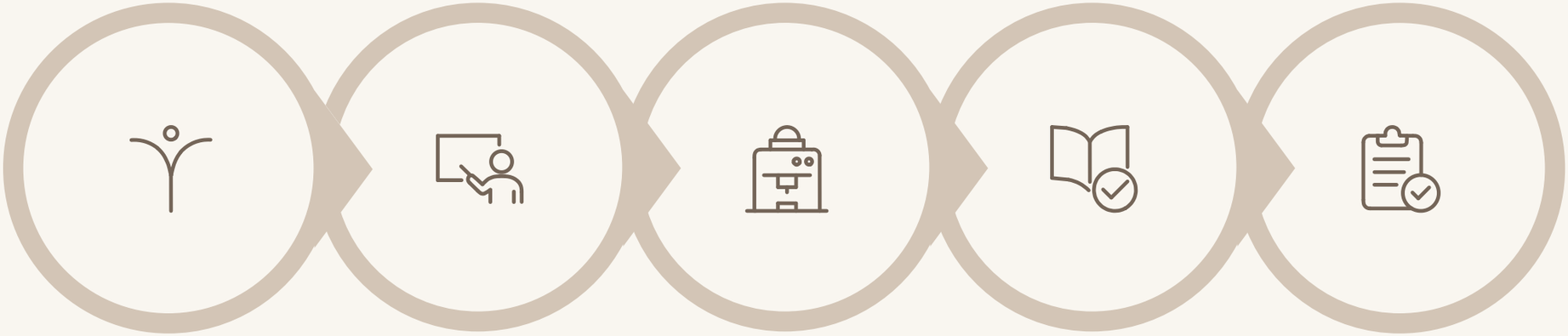
We split our dataset into **80% training data** (1,780 documents) to teach the model, and **20% test data** (445 documents) to evaluate performance on unseen examples.

Why Logistic Regression?

It's simple, fast, and works well for text classification. The model is interpretable (we can see which words influence predictions), trains quickly, and often achieves high accuracy on text data.

Multi-class Classification

Logistic Regression can handle multiple categories using techniques like one-vs-rest, where the model learns to distinguish each category from all others.



Dataset Split

80% Training

Model Training

20% Testing

Model Evaluation

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Results & Model Evaluation

Our document classification model achieved impressive results, demonstrating the power of NLP techniques for automated text categorization.

96%

Accuracy Score

Correctly classified 427 out of 445 test documents

445

Test Documents

Unseen examples used for evaluation

Understanding the Metrics

Beyond overall accuracy, we need to examine **precision**, **recall**, and **F1-score** for each category to get a complete picture of model performance.

Precision

Of all documents classified as a category, how many were actually that category? High precision means few false positives.

Recall

Of all actual documents in a category, how many did we correctly identify? High recall means few false negatives.

F1-Score

The harmonic mean of precision and recall. A balanced measure that considers both metrics - ranges from 0 to 1, with 1 being perfect.

The **confusion matrix** shows where our model made mistakes. Most misclassifications occurred between similar categories like business and politics, which makes sense since these topics sometimes overlap in news coverage.

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Conclusion & Future Improvements

Key Takeaways

- Our NLP-based document classification system achieved **96% accuracy**, demonstrating that automated text classification is highly effective
- Simple techniques like TF-IDF combined with Logistic Regression can produce excellent results without complex deep learning
- NLP enables automation of tedious manual tasks, saving time and reducing human error
- The pipeline we built can be applied to other text classification problems with minimal modifications

Future Enhancements

Deep Learning Models

Experiment with neural networks like LSTM or transformers (BERT) for potentially higher accuracy

Larger Datasets

Test the approach on larger, more diverse datasets to evaluate scalability

Real-Time Processing

Implement streaming classification for documents as they arrive in real-time

> "NLP techniques have transformed document classification from a manual, time-consuming task into an automated process that can handle thousands of documents in seconds."

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Thank You

Thank you for your time, attention, and thoughtful engagement throughout this presentation. I truly appreciate the opportunity to share this journey with you.

> “The beautiful thing about learning is that no one can take it away from you.”

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